



1100 0011 0110 1001 1111 1000 0101 1010 1001 0110 1011 1101 0100 1010 0111 0000

is a one?

0 1 0 1 1 0 1 0 1 0 1 1 0 0 1 0

ones to the left

0 0 1 1 2 3 3 4 4 5 5 6 7 7 7 8

destination

0 8 1 9 10 2 11 3 12 4 13 14 5 6 15 7

1100 0110 1000 1010 0110 0100 1010 0000 0011 1001 1111 0101 1001 1011 1101 0111

extract bit

exclusive scan

destination of a zero = element index - # ones to the left

destination of a one = input size - # ones in total + # ones to the left

Parallelization approach: Assign one thread to each element
(already know how to parallelize scan, the rest is trivial)